Teacher:	Ray Joshua Balingkit	**Subject:**	Computing
Week:	2 (Term 1)	**Year:**	7
Link:		**Cover:**	

LESSON OVERVIEW

What should students know/do by the end of this week's lessons?

Students will learn to explain that computers represent data using binary digits and identify three kinds of digital content. They will also learn to convert an unsigned binary number to decimal using place values and identify bits and bytes. Through teacher demonstrations, discussion and practical activities, students will practise these skills and explain their ideas.

End-of-Week Assessment/Check: How will you know they got it?

Use teacher observation, pupil explanations and the completed practical or paper activity. Ask pupils to name both binary digits, identify three kinds of stored data and explain why a code key matters. Check one independent conversion, correct place values and an explanation that eight bits make one byte. Note pupils who need further guided practice.

LESSON SLOTS - MONDAY AND TUESDAY

Lesson	Date	Learning Objective	The Lesson Flow	Differentiation
1	31 August 2026	By the end of this lesson, students will be able to explain that computers represent data using binary digits and identify three kinds of digital content.	**H:** Compare a printed picture with a saved picture file and ask how a computer stores it. **ID:** Introduce Unit 1 Storing digital data. Explain that bits use 0 and 1 to represent numbers, text, images, sound and instructions. **WD:** Use a supplied code key to decode a short word and show how the same bits need an agreed interpretation. **YD:** Pupils sort examples of digital content and encode a short route using a four-instruction code key. **C:** Ask pupils to name both binary digits, identify three kinds of stored data and explain why a code key matters.	**Support:** Provide place-value grids, carry rows, a character-code table and labelled diagrams. Model one step at a time and allow spoken reasoning before writing. **Extension:** Ask pupils to justify a method, identify a deliberate error or test an eight-bit boundary, then verify the result using another representation.
2	1 Septemb er 2026	By the end of this lesson, students will be able to convert an unsigned	**H:** Show 11 in decimal and binary and ask whether they have the	**Support:** Provide place-value grids, carry rows, a character-code table and labelled

		binary number to decimal using place values and identify bits and bytes.	same value. **ID:** Explain base 2 place values and that a byte contains eight bits; read binary digits separately. **WD:** Convert 00001101 to 13 by adding 8, 4 and 1, then show why leading zeros preserve the value. **YD:** Pupils use place-value grids to convert four-bit examples, then selected eight-bit examples and order the results. **C:** Check one independent conversion, correct place values and an explanation that eight bits make one byte.	diagrams. Model one step at a time and allow spoken reasoning before writing. **Extension:** Ask pupils to justify a method, identify a deliberate error or test an eight-bit boundary, then verify the result using another representation.
3				Support: Extension:
4				Support: Extension:

MATERIALS NEEDED FOR THE WEEK

- Oxford International Lower Secondary Computing Student Book 7 printed pages 4-13
- Teacher computer and display
- Classroom computers or laptops and the class save folder
- Binary place-value grids, carry rows, character-code tables and pixel or sampling sheets
- Whiteboard and markers
- Student notebooks, pencils and task sheets
- Prepared examples and the lesson success checklist



END-OF-WEEK REFLECTION

- What worked well?

- What flopped or confused students?

- What needs to be re-taught or adjusted next week?